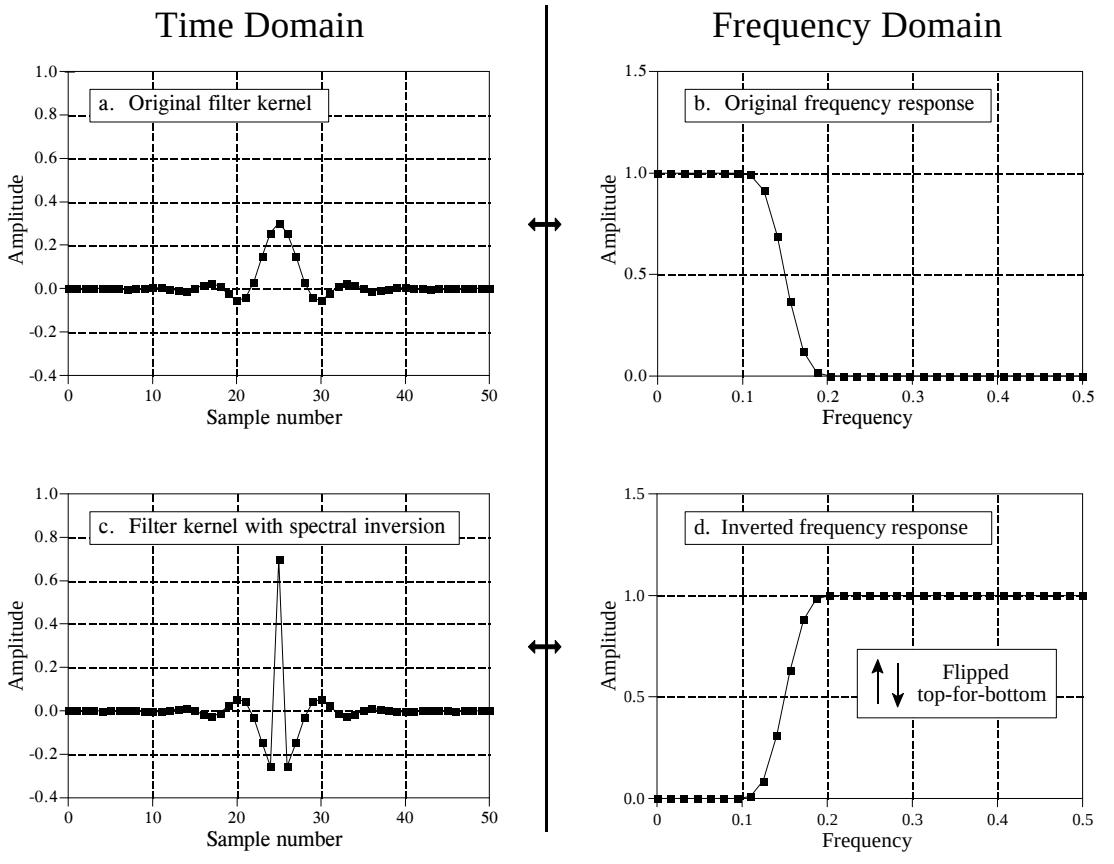

FIGURE 14-5
Example of spectral inversion. The low-pass filter kernel in (a) has the frequency response shown in (b). A high-pass filter kernel, (c), is formed by changing the sign of each sample in (a), and adding one to the sample at the center of symmetry. This action in the time domain *inverts* the frequency spectrum (i.e., flips it top-for-bottom), as shown by the high-pass frequency response in (d).

bound to have trade-offs, and might go against the conventional wisdom of this chapter. Remember the number one rule of education: *A paragraph in a book doesn't give you a license to stop thinking*.

High-Pass, Band-Pass and Band-Reject Filters

High-pass, band-pass and band-reject filters are designed by starting with a low-pass filter, and then converting it into the desired response. For this reason, most discussions on filter design only give examples of low-pass filters. There are two methods for the low-pass to high-pass conversion: **spectral inversion** and **spectral reversal**. Both are equally useful.

An example of *spectral inversion* is shown in 14-5. Figure (a) shows a low-pass filter kernel called a windowed-sinc (the topic of Chapter 16). This filter kernel is 51 points in length, although many of samples have a value so small that they appear to be zero in this graph. The corresponding